Fire Fighting and Plant Health Detection using Robot Car

Intelligent and Autonomous Systems

Murugan Chandrasekhar Omesh Sai Chethan

IIIT Sricity

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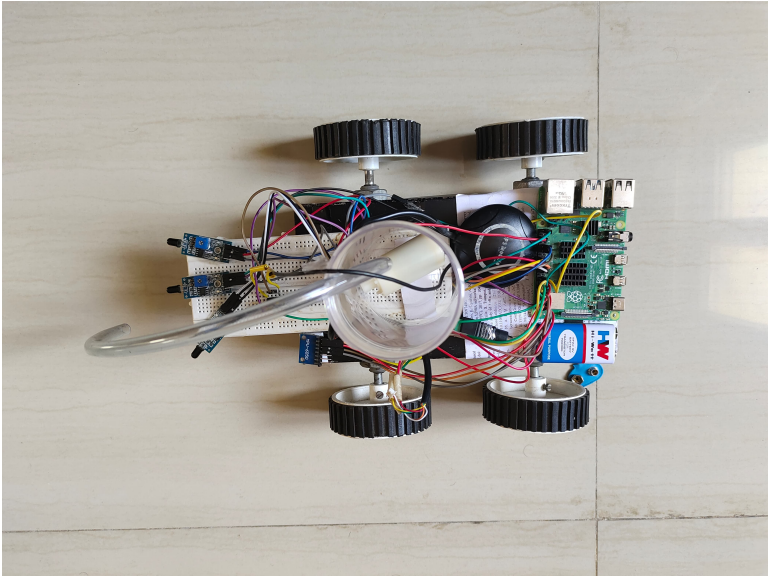
Problem Statement

- This project involves the development of a robotic car for Fire Fighting.
- System is equipped with fire detection and suppression capabilities and also monitors plant health.
- System Utilizes image processing and environmental sensors for assessment.

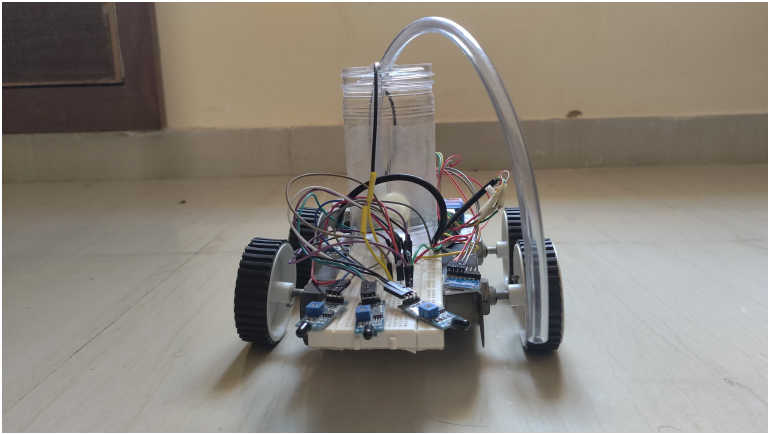
Motivation

- Fires pose a major threat to agricultural fields , forests, requiring fast and efficient suppression.
- Both fire outbreaks and environmental factors affecting plant health need early detection to prevent damage.
- Firefighting are dangerous thus an autonomous system enhances safety.
- Integrating smart sensors,and robotics enables intelligent decision-making for fire control and health diagnostics.
- A multi-functional robotic system improves emergency response and environmental monitoring, ensuring a safer ecosystem.

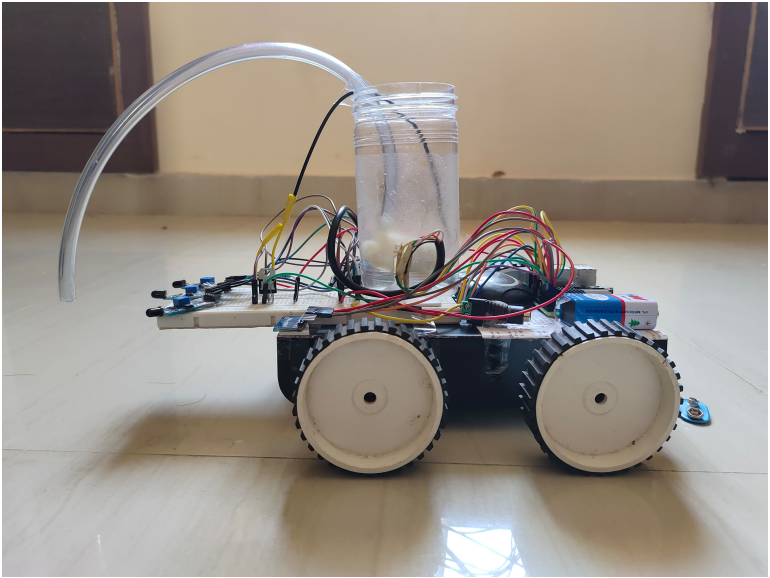
Top View of the Robot Car



Front View of the Robot Car



Side View of the Robot Car



Technical Details: Hardware Components

Hardware Components

- IR 4-pin Flame Sensors $\times 3$
- Raspberry Pi 4 Model B
- BO Motors $\times 4$ (with wheels)
- L298 Motor Driver
- Breadboard
- 5-9V Water Pump with Pipe
- Water Tank / Bottle
- IMU
- GPS
- 9V DC Battery
- Jumper Wires
- TIP-122 Transistor, 1K Resistor

Technical Details: Software & Algorithms

Software & Algorithms

- Used Raspberry Pi OS in SD Card 16GB
- Fire detection using 3 IR flame sensors data processed with Python.
- Real-time plant health analysis using a phone camera and OpenCV for Plant Monitoring.
- IMU sensor and GPS are used for reliability.

Proposed Solution

- The robotic car remains stationary (in rest mode) when there is no fire detected.
- The car uses three flame sensors (left, front, right) to detect the direction of fire. Based on which sensor is triggered, the car intelligently navigates towards the source of the fire by turning in the appropriate direction.
- Once the car determines the direction of the fire,it moves closer and activates a water pump to extinguish the fire.

Proposed Solution

- The IMU tracks the car's orientation and movement, while the GPS ensures precise location tracking, helping the robot navigate effectively during both fire detection and plant monitoring tasks..
- The robot uses a Phone camera and image processing to monitor plant health. Healthy green leaves are marked with a green boundary, while yellow or discolored leaves, indicating poor health, are highlighted with a red boundary..

limitations

- The robot cannot perform all its functions simultaneously (e.g., distance measurement, obstacle detection, and path following) due to power limitations.
- Ultrasonic sensor is needed for detecting obstacles, and maintaining a safe distance from the fire while extinguishing it. However, due to power limitations, we are unable to use it.

Thank You